

Cohort Profile: The MD Anderson 24K Clonal Hematopoiesis (MACH24) Cohort

Key Features

- MACH24 links blood sequencing with longitudinal clinical records for adults receiving care at The University of Texas MD Anderson Cancer Center. It supports studies of clonal hematopoiesis across the observed variant allele fraction (VAF) spectrum.
- The analytic cohort includes 24,126 adults aged ≥ 18 years from a sequencing denominator of 24,140 individuals, after excluding 14 participants younger than 18 years.
- Blood samples were collected from 5 August 2003 to 12 March 2024. The cohort was followed through recorded death or last-contact dates, with a median observed follow-up of 5.71 years (range, 0.05–18.54 years).
- Linked records include mortality and last contact, cancer diagnoses, treatment, radiotherapy, procedures, medications, medical history, encounters and repeated vital measurements. These records support overall survival analysis. Disease-specific outcomes require definitions and validation before analysis.
- Baseline sequencing identified at least one retained CH-associated variant in 9,904 of 24,126 participants (41.1%). Low-VAF CH only was detected in 6,203 (25.7%), conventional CHIP in 3,701 (15.3%), and no retained CH variant in 14,222 (58.9%). Among those with conventional CHIP, 1,616 also carried variants below 2% VAF (Figure 2).
- Collaborations are encouraged, and scientists who are interested in the data are welcomed to contact the corresponding author via e-mail to [KTNead@mdanderson.org].

Why was the cohort set up?

Clonal hematopoiesis (CH) arises when a hematopoietic stem or progenitor cell acquires a somatic mutation and its descendants expand. Large sequencing studies showed that detectable

CH becomes more common with age and is associated with later hematological malignancy and all-cause mortality.^{1,2} Studies have also linked CH to atherosclerotic cardiovascular disease.³ Most epidemiological and clinical studies define clonal hematopoiesis of indeterminate potential (CHIP) using a variant allele fraction (VAF) of at least 2%. The threshold was introduced as a practical working definition, not as a biological boundary.⁴ Error-corrected and ultra-sensitive sequencing studies have since detected much smaller clones throughout adult life, some of which persist over time.^{5,6} Detecting a small clone, however, does not establish that it has the prognosis or clinical implications of conventional CHIP.

CH is common in patients with non-hematological cancers and varies with age, smoking and prior therapy.⁷ Radiation, platinum compounds and topoisomerase II inhibitors may favour the growth of clones. This selection often involves DNA-damage-response genes.⁸ Published studies of CH in patients with solid tumors have used different sequencing assays, gene panels, sequencing depths and VAF thresholds. These differences limit direct comparisons of mutation prevalence and low-VAF detection across studies. A large longitudinal oncology cohort with high-depth sequencing can address this gap by assessing the clinical relevance of variants below 2% VAF.^{9,10} Linking sequencing data to clinical records allows these associations to be examined in relation to clone characteristics, patient factors and treatment exposure. The MD Anderson 24K Clonal Hematopoiesis (MACH24) Cohort was established for this purpose.

The MACH24 Cohort links blood-based CH profiling with longitudinal clinical information from adults treated at a comprehensive cancer centre. We aim to characterize CH across the observed VAF spectrum and describe differences by age, sex, ancestry, smoking status, tumor context and treatment exposure. Planned longitudinal studies will assess subsequent mortality,

hematological malignancy and other clinical outcomes using prespecified designs. The cohort also provides a resource for collaborative clinical and genomic research.

This Cohort Profile describes the rationale, cohort construction, available measurements, follow-up, and baseline genomic characteristics of MACH24. It also quantifies the mutation-positive population detected below the conventional 2% CHIP threshold. The baseline analyses do not estimate the clinical consequences of low-VAF CH.

Who is in the cohort?

MACH24 is a hospital-based cohort of patients who received care at The University of Texas MD Anderson Cancer Center and had available blood samples, targeted CH sequencing data and linked institutional clinical records. Cohort enrolment is closed, but follow-up of existing participants is ongoing through institutional clinical records.

The sequencing denominator contained 24,140 unique participants. Eligibility required presence in this denominator and age at blood collection of at least 18 completed years. Participants were linked to clinical records by normalized medical record number, with sample and laboratory identifiers retained for cross-checking. Age was calculated using full-year birth dates from the demographic records and the index blood-collection date. Fourteen participants were younger than 18 years, leaving 24,126 adults (Figure 1). No additional exclusion based on hematological diagnoses, missing diagnosis dates or billing records was applied in this release.

The median age at blood collection was 64.1 years [interquartile range (IQR), 57.0–70.6], and 12,150 (50.4%) were male. Recorded race was White in 19,386 (80.4%), Black in 1,942 (8.0%), Asian in 747 (3.1%), Other in 1,503 (6.2%), Unknown in 548 (2.3%). Smoking status was never in 9,730 (40.3%), ever in 8,124 (33.7%), unknown in 6,272 (26.0%) (Table 1). The age distribution is shown in Supplementary Figure S1.

Recorded primary solid tumors include breast, prostate, lung, colorectal, gynaecological, kidney, head and neck, bladder, thyroid and pancreatic cancers. Tumor-site assignments come primarily from Cancer Registry records. Other clinical diagnosis records provide a fallback when registry information is unavailable. For participants with multiple primary tumors, rules for assigning tumor sites must be specified before tumor-specific causal or prognostic analyses.

How often have participants been followed up?

The index date was the date of blood collection for CH assessment. Blood samples were collected from August 5, 2003 through March 12, 2024. Cohort enrolment is closed, but follow-up continues through institutional clinical records rather than scheduled research visits.

Follow-up was measured from the index blood draw to the verified date of death or, for participants without a recorded death, the last recorded contact. A valid observed endpoint was available for 24,125 of 24,126 participants. Median observed follow-up was 5.71 years (range, 0.05–18.54 years), with 8,280 verified deaths after blood collection. Dates were evaluated against a data-review cutoff of 16 September 2026. The participant without a valid endpoint was omitted from survival estimation; no contact date was imputed.

Longitudinal data available through the Foundry platform include dated encounters, neoplastic diagnoses, systemic treatment summaries, radiotherapy, medications, procedures, medical history and repeated vital measurements. Recording frequency varies with clinical care and the type of measurement. Events occurring outside MD Anderson may not be fully captured.

What has been measured?

Clonal hematopoiesis profiling

The cohort includes high-depth targeted sequencing data generated from stored blood samples using a custom CH panel and variant-calling pipeline. The assay was based on the method described by Mack et al., with higher sequencing depth and the addition of CHEK2 to the published 22-gene panel. Supplementary Table S1 summarizes the capture intervals and non-overlapping target lengths for these 23 genes. Supplementary Table S2 provides the complete capture-design coordinates and annotations in hg38. All samples included in the analysis passed laboratory sequencing quality control.

The variant-calling annotations identified Mutect2 and VarDict, with additional calls attributed to VarScan2 and Pindel. For the primary analysis, we excluded variants flagged for fewer than 12 alternate-supporting reads, suspicious recurrence or occurrence in the panel of normals. These exclusions used the quality flags supplied with the variant data. Variant allele fraction (VAF) was defined as the proportion of reads supporting the alternate allele at a given site. The lower VAF threshold for the primary analysis was 0.3%; this threshold defined analytical inclusion rather than an independently validated assay limit of detection.

Participants were assigned to three mutually exclusive categories according to their retained CH-associated variants. Low-VAF CH was defined as at least one variant with $\text{VAF} \geq 0.3\%$ but $< 2\%$, with no variant at $\text{VAF} \geq 2\%$. Conventional clonal hematopoiesis of indeterminate potential (CHIP) was defined operationally as at least one retained variant with $\text{VAF} \geq 2\%$. Participants in this category could also carry variants below 2% VAF. Participants with no variant meeting the analysis criteria were classified as having no detected CH; this designation does not exclude clones below the analytical threshold or variants outside the assessed regions.

Participant-level CH categories were distinguished from variant-level VAF strata in the gene-specific analyses. Each retained variant was assigned to the VAF <2% or VAF $\geq$ 2% stratum according to its own VAF. Thus, a participant carrying variants in both strata contributed to both variant-level summaries but was classified only in the conventional CHIP group for participant-level analyses.

Baseline characteristics and exposures

Baseline characteristics included age at blood collection, sex, recorded race and smoking status. Age was calculated from the dates of birth and blood collection. Sex, race and smoking status were obtained from institutional clinical records. Smoking status was classified as never, ever or unknown. Missing race and smoking information were retained as an unknown category in the descriptive analyses (Table 1).

Primary solid-tumor diagnoses and stage were obtained from the Cancer Registry and other clinical diagnosis records, with registry records serving as the primary source. Linked records also contained medical history, systemic treatment, radiotherapy, medications, procedures, encounters and repeated vital measurements (Table 2). Dated records allow exposures before blood collection to be distinguished from treatments and clinical events recorded during follow-up.

Longitudinal outcomes

Overall survival was defined as the time from the index blood draw to death from any cause. Verified death dates were used to identify deaths, and participants without a recorded death were censored at their last recorded contact.

The Cancer Registry contained records of hematological malignancy dated after blood collection for 126 participants. These records identify potential incident cases. However, the registry date may represent either the diagnosis date or the first recorded contact for the malignancy.

Confirmation of disease onset is therefore needed to distinguish new diagnoses from previously existing diseases.

Linked clinical records also contain information on subsequent solid tumors, metastatic disease, cardiovascular diagnoses and other complications. These data provide a basis for studying additional outcomes, although a diagnosis recorded after blood collection does not by itself establish a new event, cancer progression or recurrence.

What has the cohort found? Key findings and publications

Baseline sequencing identified at least one retained CH-associated variant in 9,904 of 24,126 participants (41.1%). Low-VAF CH only was detected in 6,203 (25.7%), conventional CHIP in 3,701 (15.3%), and no retained CH variant in 14,222 (58.9%). Among those with conventional CHIP, 1,616 also carried variants below 2% VAF (Figure 2).

DNMT3A and TET2 were the most frequently affected genes in both VAF classes (0.3% to <2% and $\geq 2\%$). Variants in PPM1D, CHEK2, TP53 and ASXL1 were also common (Figure 3). Low-VAF variants had a median sequencing depth of $2,972.5\times$ (IQR, 2,467–3,516) and a median alternate-read count of 21 (IQR, 16–31). Sensitivity analyses using alternative variant filters and VAF thresholds consistently identified a substantial group with variants below 2%, although its size varied with the criteria applied.

Both low-VAF CH only and conventional CHIP were more common at older ages. In separate logistic models adjusted for age, sex, race and smoking status, each category was compared with no detected CH. Per 10-year increase in age, adjusted odds ratios were 1.66 (95% CI, 1.61–1.72)

for low-VAF CH only and 2.21 (95% CI, 2.12–2.31) for conventional CHIP. For ever versus never smoking, corresponding estimates were 1.10 (95% CI, 1.03–1.19) and 1.27 (95% CI, 1.16–1.39) (Supplementary Figure S2). These are baseline associations, not causal effects.

The predominance of DNMT3A and TET2 and the association with age are consistent with earlier population studies.^{1,2,5,6} PPM1D, TP53 and CHEK2 have also been reported in oncology cohorts, where cancer therapy may select for particular clones.^{7,8} In MACH24, approximately one-quarter of participants had CH-associated variants exclusively below the conventional CHIP threshold. Their clinical significance remains to be established through longitudinal outcome analyses.

During follow-up, 8,280 deaths were recorded among 24,125 participants with valid endpoints. Unadjusted Kaplan–Meier curves generally showed lower survival in the conventional CHIP group than in the group with no detected CH, with the low-VAF-only group intermediate over most of follow-up (Supplementary Figure S7). These descriptive comparisons do not account for differences in age, cancer type, stage or treatment.

What are the main strengths and weaknesses?

MACH24 combines a large cohort of oncology patients, high-depth blood sequencing and longitudinal clinical data. Detection of variants below the conventional 2% VAF threshold allows low-VAF CH and conventional CHIP to be examined within the same cohort. The broad range of tumor types and linked treatment histories support studies of CH across different oncology settings. A median observed follow-up of 5.71 years provides a basis for evaluating associations between baseline CH and subsequent clinical outcomes.

Eligibility in this release was restricted by sequencing-denominator membership and adult age, without hematological malignancy exclusions or clinical adjudication of all CH categories.

Conventional CHIP is therefore an operational VAF-based category rather than confirmation that every carrier meets all clinical diagnostic criteria for CHIP. Studies requiring absence of a pre-existing hematological malignancy need additional disease-specific eligibility assessment.

The cohort has several limitations. Patients were recruited from a single cancer center, and inclusion depended on the availability of blood samples and linked clinical data. Referral patterns, tumor distribution and biospecimen availability may limit generalizability to other oncology populations. Survival to blood collection may introduce selection bias, with rapidly fatal disease potentially underrepresented. Treatment received before blood collection may also have influenced the observed clonal profiles, while the timing of clone emergence remains unknown.

Follow-up relies on routine clinical records, with variation in the frequency and completeness of measurements across participants. Events occurring outside MD Anderson may not be captured. In addition, a diagnosis first recorded after blood collection does not necessarily indicate new disease onset. Analyses of disease-specific outcomes require review of diagnosis dates and supporting clinical evidence to distinguish incident from pre-existing disease.

Detection at very low VAF depends on assay performance and variant-filtering criteria. High sequencing depth alone does not establish analytical sensitivity or eliminate artefacts, and differences between assays may affect comparisons across studies. Baseline sequencing provides a single assessment of CH. Serial sampling is needed to determine whether individual clones persist, expand or disappear.

Can I get hold of the data? Where can I find out more?

The cohort contains individual-level genomic and clinical data from patients treated at MD Anderson Cancer Center. These data are sensitive and are not currently available as an unrestricted public dataset.

Aggregate definitions, analysis code and non-identifying source data underlying published figures may be shared subject to institutional policy and manuscript-specific approval.

Profile in a nutshell

Characteristic	MACH24 cohort
Cohort type	Longitudinal single-centre clinical-genomic cohort
Setting	The University of Texas MD Anderson Cancer Center
Source cohort	24,140 participants in the sequencing denominator
Current analytic cohort	24,126 adults aged ≥ 18 years at blood collection
Index period	5 August 2003 to 12 March 2024
Follow-up	Median 5.71 years; range 0.05–18.54 years
Follow-up completeness	24,125/24,126 with a valid death or last-contact endpoint
Baseline genomic exposure	Targeted blood-based CH sequencing; all included participants are on the sequencing denominator
Current CH categories	No retained CH variant detected; low- <i>VOF</i> CH only; conventional CHIP
Analysis-ready outcome	Overall survival from blood draw
Candidate outcomes requiring validation	Incident hematological malignancy, cardiovascular outcomes, subsequent solid-tumor events and treatment-related complications
Data access	Controlled; governance and access procedure to be specified

Ethics

Funding

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Author contributions

Conflict of interest

Data availability

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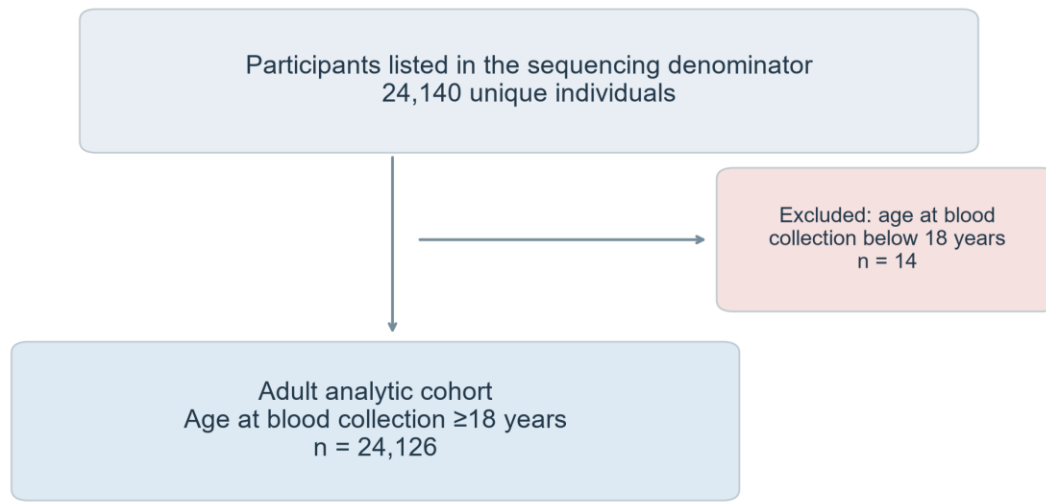
Tables

Table 1. Baseline characteristics by clonal haematopoiesis category. Values are n (%) unless otherwise indicated. Categories are mutually exclusive at participant level.

Characteristic	No detected CH	Low-VAF CH only	Conventional CHIP	Overall
Age, years, mean (SD)	61.2 (10.5)	66.2 (9.1)	68.8 (9.0)	63.7 (10.4)
Age, years, median [IQR]	61.4 [55.0, 68.1]	66.4 [59.8, 72.4]	68.9 [62.7, 75.1]	64.1 [57.0, 70.6]
Male sex	7085 (49.8%)	3155 (50.9%)	1910 (51.6%)	12150 (50.4%)
Race: White	11249 (79.1%)	5043 (81.3%)	3094 (83.6%)	19386 (80.4%)
Race: Black	1171 (8.2%)	503 (8.1%)	268 (7.2%)	1942 (8.0%)
Race: Asian	465 (3.3%)	190 (3.1%)	92 (2.5%)	747 (3.1%)
Race: Other	983 (6.9%)	336 (5.4%)	184 (5.0%)	1503 (6.2%)
Race: Unknown	354 (2.5%)	131 (2.1%)	63 (1.7%)	548 (2.3%)
Smoking: Never	6126 (43.1%)	2345 (37.8%)	1259 (34.0%)	9730 (40.3%)
Smoking: Ever	4551 (32.0%)	2164 (34.9%)	1409 (38.1%)	8124 (33.7%)
Smoking: Unknown	3545 (24.9%)	1694 (27.3%)	1033 (27.9%)	6272 (26.0%)

Table 2. Clinical-genomic measurement domains and longitudinal availability. Clinical observation is encounter-driven rather than based on fixed research visits. Outcome-specific definitions and temporal windows must be prespecified before inferential analyses.

Domain	Source	Timing or repeatability	Current use
Index blood draw	Biospecimen record	One index date; 2003-2024	Baseline and time zero
CH variants	Targeted blood sequencing	Baseline; one participant-level CH classification	CH classification and gene/VAF profile
Demographics and smoking	Clinical records	Baseline or closest prespecified window	Description and covariate adjustment
Cancer diagnosis and stage	Cancer Registry and clinical diagnosis records	Before and after blood draw	Tumor context and candidate outcomes
Cancer treatment	Treatment, medication, procedure and radiotherapy records	Dated longitudinal events	Time-varying or windowed exposures
Mortality and contact	Verified death and last-contact dates	Evaluated through 16 September 2026	Overall survival and censoring
Hematological malignancy	Cancer Registry and diagnosis records	After blood draw	Candidate incident outcome requiring phenotype finalization
Other clinical outcomes	Diagnoses, procedures, medications, encounters and vital measurements	Longitudinal and encounter-driven	Candidate phenotypes requiring validation



Eligibility uses the denominator and full-year birth dates.
No hematologic-diagnosis or billing exclusion was applied.

Figure 1. Cohort assembly. The sequencing denominator included 24,140 unique individuals. Excluding 14 participants younger than 18 completed years at blood collection left 24,126 adults. Full-year demographic birth dates were used to calculate age. Denominator membership and adult age were the only participant-level eligibility criteria; hematological diagnosis and billing rules were not applied.

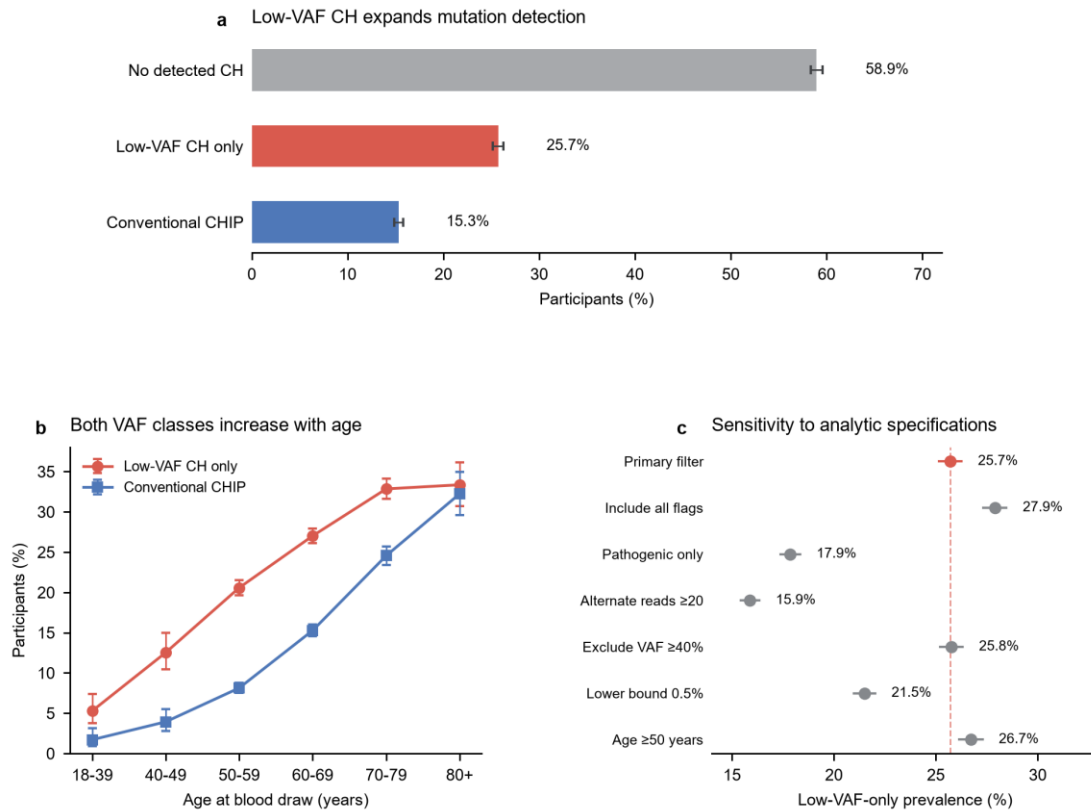


Figure 2. Detection of low-VAF clonal hematopoiesis. a, Mutually exclusive categories among 24,126 participants: no detected CH, 14,222 (58.9%); low-VAF CH only, 6,203 (25.7%); and conventional CHIP, 3,701 (15.3%). Low-VAF CH only requires at least one retained variant with $0.3\% \leq \text{VAF} < 2\%$ and none with $\text{VAF} \geq 2\%$; conventional CHIP requires a retained variant with $\text{VAF} \geq 2\%$. b, Category prevalence by age at collection. Denominators were 586, 814, 7,094, 9,118, 5,351, 1,163 for ages 18–39, 40–49, 50–59, 60–69, 70–79 and ≥ 80 years. c, Low-VAF-only prevalence under the primary definition and six alternative specifications. The age ≥ 50 analysis included 22,726 participants; other displayed specifications included 24,126. Error bars show two-sided 95% Wilson confidence intervals.

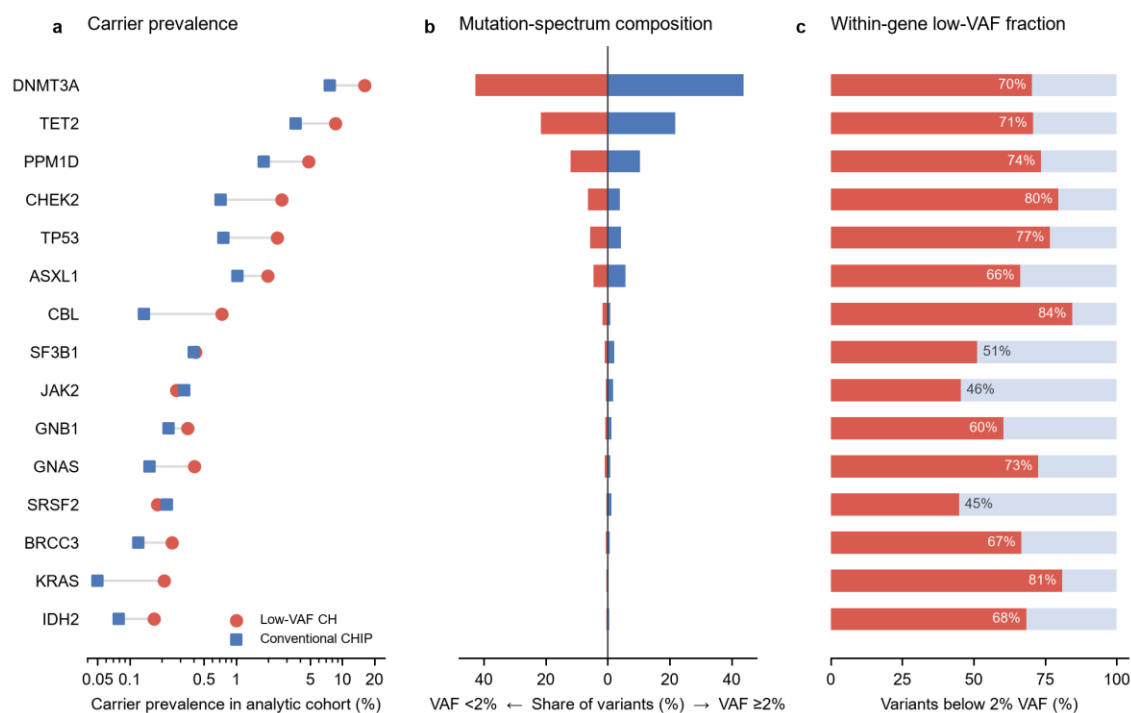


Figure 3. Gene-level landscape of low-VAF clonal hematopoiesis and conventional CHIP. a, Gene-specific carrier prevalence among 24,126 participants for the 15 genes with the largest retained-variant counts; the horizontal axis is logarithmic. A participant can contribute to both VAF strata. b, Gene composition within the 10,840 variants with VAF<2% (left) and 4,434 variants with VAF≥2% (right). c, Percentage of each gene's retained variants below 2% VAF. These are descriptive summaries; no gene-to-gene hypothesis tests were performed.